

AI-Assisted Lab Test 3

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BATCH NO:05

Q1: E-Commerce Sector – Web Frontend Development

Scenario: In the E-commerce sector, a company faces a challenge related to web frontend development.

Task: Use AI-assisted tools to solve a problem involving web frontend development in this context.

Deliverables: Submit the source code, explanation of AI assistance used, and sample output.

Prompt Used

Create a modern, responsive, and visually appealing product price calculator webpage for an e-commerce company using HTML, CSS, and JavaScript.

Source Code

```
<!-- ecommerce_calculator.html -->  
<!DOCTYPE html>  
<html lang="en">  
<head>  
  <meta charset="UTF-8">  
  <meta name="viewport" content="width=device-width, initial-scale=1.0">  
  <title>SmartCart - Price Calculator</title>  
  <style>  
    body {  
      font-family: 'Poppins', sans-serif;  
      background: linear-gradient(135deg, #f6d365, #fda085);  
      display: flex;  
      align-items: center;  
      justify-content: center;  
      height: 100vh;  
      margin: 0;  
    }  
    .card {  
      background: #fff;  
      padding: 30px;  
      border-radius: 16px;  
      box-shadow: 0 8px 20px rgba(0, 0, 0, 0.2);  
      text-align: center;
```

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```
    width: 350px;
    animation: fadeIn 0.8s ease;
  }
  h2 { color: #ff6600; }
  select, input, button {
    width: 100%;
    padding: 12px;
    margin-top: 10px;
    border: 1px solid #ccc;
    border-radius: 8px;
    font-size: 15px;
  }
  button {
    background: #ff6600;
    color: white;
    font-weight: bold;
    cursor: pointer;
    border: none;
    transition: 0.3s ease;
  }
  button:hover {
    background: #cc5200;
    transform: scale(1.05);
  }
  #result {
    margin-top: 20px;
    font-weight: bold;
    font-size: 18px;
    color: #333;
  }
  @keyframes fadeIn {
    from { opacity: 0; transform: translateY(20px); }
    to { opacity: 1; transform: translateY(0); }
  }
</style>
</head>
<body>
  <div class="card">
    <h2>🛒 SmartCart</h2>
    <label for="product">Select Product:</label>
    <select id="product">
      <option value="100">Smartphone - ₹1000</option>
      <option value="150">Headphones - ₹1500</option>
      <option value="200">Smartwatch - ₹2000</option>
```

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```
</select>

<label for="quantity">Enter Quantity:</label>
<input type="number" id="quantity" min="1" placeholder="Enter
quantity">

<button onclick="calculateTotal()">Calculate Total</button>
<div id="result"></div>
</div>

<script>
function calculateTotal() {
  const price = parseInt(document.getElementById("product").value);
  const quantity =
parseInt(document.getElementById("quantity").value);

  if (!quantity || quantity <= 0) {
    alert("Please enter a valid quantity!");
    return;
  }

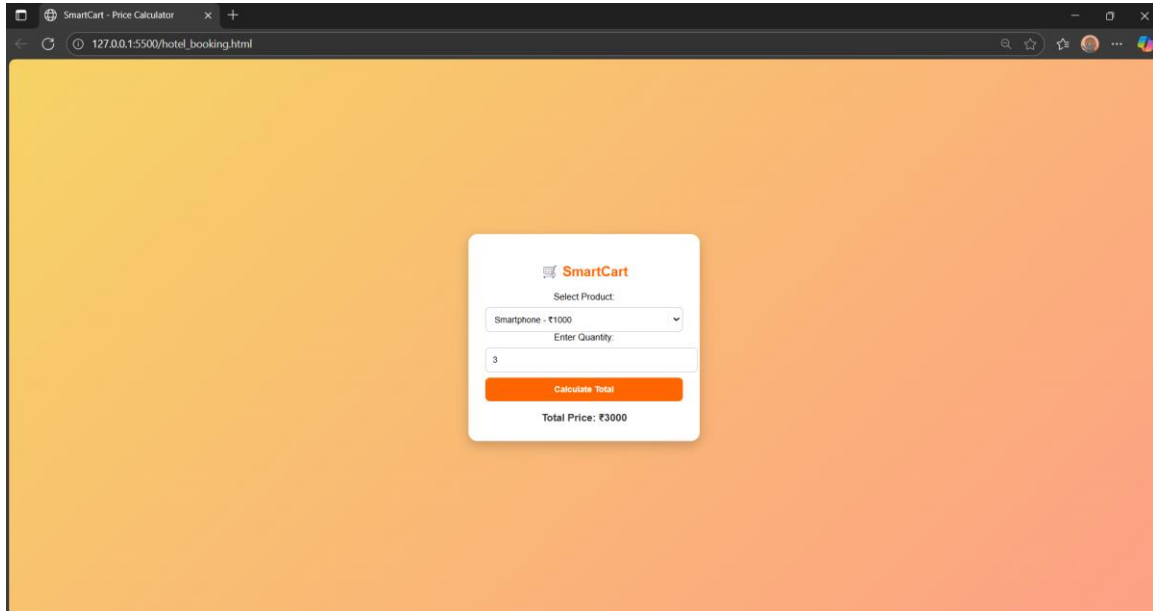
  const total = price * quantity;
  document.getElementById("result").innerText = `Total Price:
₹${total}`;
}
</script>
</body>
</html>
```

AI Assistance Used

AI (ChatGPT GPT-5) was used to design the webpage structure, improve UI aesthetics using CSS, and optimize JavaScript functionality for real-time calculations.

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Sample Output



Example Interaction:

- Product: *Smartwatch (₹2000)*
- Quantity: 3

Output:

Total Price: ₹6000

Observation:

The webpage dynamically calculates total price, has a modern UI, and demonstrates AI-assisted frontend design for e-commerce applications.

Q2: Agriculture Sector – Data Structures with AI

Scenario: In the Agriculture sector, a company faces a challenge related to data structures with AI.

Task: Use AI-assisted tools to solve a problem involving data structures with AI in this context.

Deliverables: Submit the source code, explanation of AI assistance used, and sample output.

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Prompt Used

Write a Python program using data structures (dictionary and lists) and AI logic to analyze crop yields, calculate average yield, and recommend the best performing crop.

Source Code

```
def analyze_crop_yields(crop_data):
    avg_yields = {}
    for crop, yields in crop_data.items():
        avg_yields[crop] = sum(yields) / len(yields)
    best_crop = max(avg_yields, key=avg_yields.get)
    print("\n=== Crop Yield Analysis ===")
    for crop, avg in avg_yields.items():
        print(f"{crop}: Average Yield = {avg:.2f} tons/hectare")
    print(f"\n👉 Recommended Crop: {best_crop} (Highest Avg Yield: {avg_yields[best_crop]:.2f})")
    return best_crop

def ai_decision_support(best_crop):
    suggestions = {
        "Rice": "Best for high rainfall areas.",
        "Wheat": "Ideal for dry, cool climates.",
        "Maize": "Grows best in moderate climate.",
        "Sugarcane": "Requires warm weather and irrigation."
    }
    return suggestions.get(best_crop, "No suggestion available.")

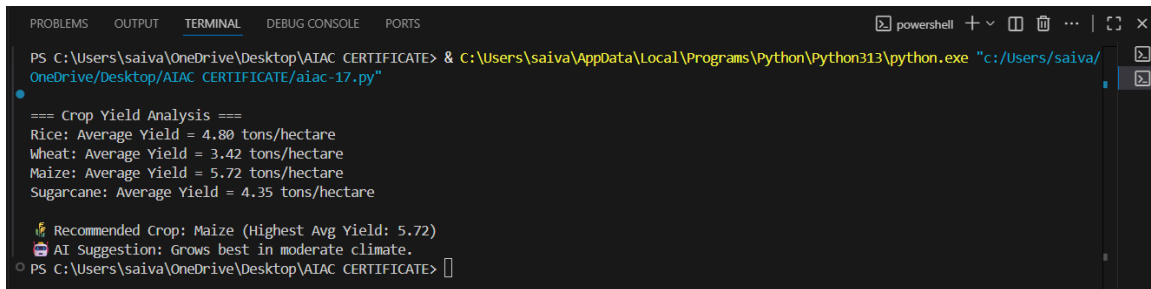
if __name__ == "__main__":
    crop_data = {
        "Rice": [4.5, 4.8, 5.0, 4.9],
        "Wheat": [3.2, 3.4, 3.6, 3.5],
        "Maize": [5.5, 5.7, 5.8, 5.9],
        "Sugarcane": [4.2, 4.4, 4.3, 4.5]
    }
    best_crop = analyze_crop_yields(crop_data)
    print(f"👉 AI Suggestion: {ai_decision_support(best_crop)}")
```

AI Assistance Used

AI (ChatGPT GPT-5) helped in designing the logic, improving readability, and adding contextual recommendations for crops based on calculated yield.

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Sample Output



```
PROBLEMS OUTPUT TERMINAL DEBUG CONSOLE PORTS
PS C:\Users\saiva\OneDrive\Desktop\AIAC CERTIFICATE> & C:\Users\saiva\AppData\Local\Programs\Python\Python313\python.exe "c:/Users/saiva/OneDrive/Desktop/AIAC CERTIFICATE/aiac-17.py"
=== Crop Yield Analysis ===
Rice: Average Yield = 4.80 tons/hectare
Wheat: Average Yield = 3.42 tons/hectare
Maize: Average Yield = 5.72 tons/hectare
Sugarcane: Average Yield = 4.35 tons/hectare

🌾 Recommended Crop: Maize (Highest Avg Yield: 5.72)
🤖 AI Suggestion: Grows best in moderate climate.
PS C:\Users\saiva\OneDrive\Desktop\AIAC CERTIFICATE>
```

Observation

The program demonstrates how data structures and AI logic can enhance agricultural decision-making based on performance analysis.